Automated Communications Research Paper

Automated LLM Agent System

***Abstract*—Massive connectivity in Cell-Free MIMO systems is significantly constrained by limited fronthaul capacity, where traditional fixed-quantization schemes often fail to adapt to dynamic channel conditions. To address this, we propose a Task-Oriented Resource-Inference Graph Neural Network (GNN) framework that jointly optimizes bit-width allocation and signal detection. Unlike conventional methods that separate quantization from detection, our approach deploys a lightweight, local Policy Network at each Access Point (AP) to infer the optimal quantization bit-width (0, 2, or 4 bits) based on instantaneous local Channel State Information (CSI), coupled with a central Bit-Aware GNN for robust signal detection. The framework leverages Gumbel-Softmax relaxation to enable differentiable discrete bit selection and employs Learned Step Size Quantization (LSQ) to minimize quantization error during training. Numerical results demonstrate that the proposed system achieves detection performance comparable to full-precision 32-bit baselines while consuming only approximately 2.5 bits on average. Furthermore, the framework exhibits enhanced robustness, maintaining a Bit Error Rate (BER) below 2.2% even when 50% of APs are disconnected, a scenario where traditional fixed-quantization schemes experience significant performance degradation.**

I. Introduction

The evolution toward the sixth generation (6G) of wireless networks is characterized by a fundamental paradigm shift from cell-centric architectures to user-centric Cell-Free Massive Multiple-Input Multiple-Output (MIMO) systems. By deploying a large number of distributed Access Points (APs) that cooperatively serve User Equipments (UEs) via a Central Processing Unit (CPU), Cell-Free MIMO offers significant theoretical gains in spectral efficiency and uniform coverage probability. However, the practical realization of these gains is severely constrained by the fronthaul capacity bottleneck. In realistic deployments, the massive volume of in-phase and quadrature (IQ) samples generated by distributed APs must be transmitted to the CPU for joint detection. Since transmitting full-precision IQ samples scales linearly with the number of antennas and bandwidth, it imposes an unsustainable load on limited-capacity fronthaul links. Consequently, the design of efficient quantization and compression schemes that reduce data volume without compromising detection accuracy has become a critical challenge.

Traditionally, fronthaul compression relies on fixed-rate quantization methods, such as uniform quantization or Lloyd-Max algorithms. While these approaches minimize the mean squared error (MSE) between the original and quantized signals, they are fundamentally agnostic to the downstream signal detection task. As noted in [1], the objective of a communication receiver should be to maximize the relevant mutual information rather than simply reconstructing the received waveform. Conventional schemes typically assign equal bit-widths to all APs regardless of their instantaneous Channel State Information (CSI), leading to inefficient resource utilization. For instance, an AP experiencing deep fading or heavy interference contributes little to the final detection but consumes the same fronthaul bandwidth as an AP with a strong line-of-sight connection. Although low-complexity detection algorithms have been analyzed for large-scale systems [2], [3], they generally assume perfect fronthaul transmission or neglect the non-linear distortion introduced by aggressive quantization.

In recent years, deep learning (DL) has emerged as a powerful tool for physical layer optimization. Model-driven approaches, such as deep unfolding, have been successfully applied to MIMO detection [4]–[6], demonstrating superior performance over traditional iterative algorithms. However, these works primarily focus on the detection algorithm at the receiver and often overlook the distributed quantization constraints inherent to Cell-Free architectures. To address bandwidth limitations, the concept of Task-Oriented Communication (TOC) has gained traction, often leveraging the Information Bottleneck (IB) principle to extract and transmit only task-relevant features [7], [8]. For example, [9] proposes semantic information recovery, while [10] applies Graph Neural Networks (GNNs) and the Graph Information Bottleneck (GIB) to reduce communication overhead in graph data. Similarly, [11] explores task-oriented channel allocation for multi-agent systems, and [12] investigates scalable edge sensing. Despite these advancements, a critical gap remains: existing TOC frameworks often separate the resource allocation decision from the signal detection process or rely on static bit-allocation strategies. There is a lack of a unified framework that allows distributed APs to dynamically infer the optimal quantization precision—including the decision to transmit zero bits—based on local CSI, while the central detector adaptively processes these heterogeneous multi-precision signals.

To bridge this gap, this paper proposes a Joint Quantization-Aware Training (QAT) framework for backhaul-constrained Cell-Free MIMO. Unlike prior works that treat quantization and detection as separate modules, we integrate a lightweight Resource Inference Module at each AP with a Heterogeneous GNN at the CPU. Our approach is grounded in the principle of task-oriented communication: the system is optimized to minimize the final symbol detection error rather than the intermediate quantization distortion. By employing a differentiable bit-selection mechanism, the network learns to allocate higher bit-widths to APs with high-quality observations and silence APs that provide negligible information, thereby maximizing detection performance under strict capacity constraints.

The main contributions of this paper are summarized as follows:

- We propose a differentiable bit-selection mechanism utilizing Gumbel-Softmax relaxation, which enables the end-to-end training of discrete quantization levels (0, 2, or 4 bits). This allows the AP-side Policy Network

to dynamically adjust the compression ratio based on instantaneous local CSI.
- We design a Bit-Aware GNN detector at the CPU that incorporates quantization precision as an attention bias. This architecture allows the CPU to robustly aggregate signals with heterogeneous quality, effectively compensating for the quantization noise introduced by low-precision APs.
- We provide extensive numerical results demonstrating the robustness of the proposed framework. The system maintains a Bit Error Rate (BER) below 2.2% even in extreme scenarios where 50% of APs are disconnected, and achieves detection performance comparable to full-precision baselines while consuming only approximately 2.5 bits per symbol on average.

The remainder of this paper is organized as follows. Section II describes the system model and the problem formulation. Section III details the proposed Joint QAT framework, including the Resource Inference Module and the Bit-Aware GNN. Section IV presents the numerical results and performance analysis. Finally, Section V concludes the paper.

II. System Model and Problem Formulation

We consider the uplink of a Cell-Free Massive MIMO system comprising L distributed Access Points (APs), indexed by the set $\mathcal{L} = \{1, \dots, L\}$, and K single-antenna User Equipments (UEs), indexed by $\mathcal{K} = \{1, \dots, K\}$. Each AP is equipped with N antennas and is connected to a Central Processing Unit (CPU) via a capacity-constrained fronthaul link. The APs cooperate to serve the users in the same time-frequency resource block. We assume a block-fading channel model where channel statistics remain constant within a coherence block.

A. Physical Layer Channel Model

Let $\mathbf{h}_{lk} \in \mathbb{C}^{N\times 1}$ denote the channel vector between the k-th user and the l-th AP. The channel models large-scale fading (including path loss and shadowing) and small-scale fading, expressed as:

$$\mathbf{h}_{lk} = \sqrt{\beta_{lk}}\mathbf{g}_{lk}, \tag{1}$$

where $\beta_{lk} \in \mathbb{R}_+$ represents the large-scale fading coefficient, which is assumed to be known at the CPU and the l-th AP due to its slow-varying nature. The vector $\mathbf{g}_{lk} \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}_N)$ represents the small-scale Rayleigh fading component.

During the uplink data transmission phase, all K users simultaneously transmit their data symbols $s_k \in \mathbb{C}$ to the APs, where $\mathbb{E}[|s_k|^2] = 1$. The received signal $\mathbf{y}_l \in \mathbb{C}^{N\times 1}$ at the l-th AP is given by:

$$\mathbf{y}_l = \sum_{k=1}^{K} \sqrt{p_k}\mathbf{h}_{lk}s_k + \mathbf{n}_l, \tag{2}$$

where p_k denotes the transmit power of the k-th user, and $\mathbf{n}_l \sim \mathcal{CN}(\mathbf{0}, \sigma^2\mathbf{I}_N)$ is the additive white Gaussian noise (AWGN) vector at the l-th AP with noise power σ^2.

B. Local Processing and Quantization

To alleviate the fronthaul burden, each AP performs local processing to extract user-specific features before transmission to the CPU. Instead of forwarding the high-dimensional raw signal $\mathbf{y}_l$, the l-th AP computes a local linear minimum mean square error (LMMSE) estimate for each user k. The local estimate $\hat{s}_{lk}$ is obtained as:

$$\hat{s}_{lk} = \mathbf{w}_{lk}^H \mathbf{y}_l, \tag{3}$$

where $\mathbf{w}_{lk} \in \mathbb{C}^{N\times 1}$ is the local combining vector. Assuming the AP has access to local Channel State Information (CSI), the optimal LMMSE combiner is given by:

$$\mathbf{w}_{lk} = p_k \left(\sum_{i=1}^{K} p_i \mathbf{h}_{li}\mathbf{h}_{li}^H + \sigma^2 \mathbf{I}_N \right)^{-1} \mathbf{h}_{lk}. \tag{4}$$

This local estimation reduces the data dimensionality from N antennas to K scalar estimates per AP. However, transmitting these continuous-valued complex scalars $\hat{s}_{lk}$ still requires infinite precision. In practical limited-backhaul scenarios, these estimates must be quantized.

We define a discrete set of admissible bit-widths $\mathcal{B} = \{0, 2, 4\}$ bits per complex symbol. The choice of $b_{lk} \in \mathcal{B}$ determines the fidelity of the signal transmitted from AP l for user k. Specifically:

- If $b_{lk} = 0$, the AP discards the signal for user k, consuming zero bandwidth.
- If $b_{lk} \in \{2, 4\}$, the AP applies a quantization function $\mathcal{Q}(\cdot; b_{lk})$ to $\hat{s}_{lk}$, generating a quantized message $\dot{s}_{lk}$.

The quantization process introduces a distortion term q_{lk}, such that the signal received at the CPU is $\dot{s}_{lk} = \hat{s}_{lk} + q_{lk}$. The statistical properties of q_{lk} depend on the chosen bit-width b_{lk} and the dynamic range of the input signal.

C. Problem Formulation

The CPU aggregates the quantized signals $\dot{\mathbf{s}} = \{\dot{s}_{lk} \mid l \in \mathcal{L}, k \in \mathcal{K}\}$ to detect the transmitted symbol vector $\mathbf{s} = [s_1, \dots, s_K]^T$. Let $f_\Theta(\cdot)$ denote the centralized detection function parameterized by Θ (e.g., a neural network), which produces the final estimate $\hat{\mathbf{s}} = f_\Theta(\dot{\mathbf{s}})$.

The core challenge is to jointly optimize the bit-width allocation $\mathbf{B} = \{b_{lk}\}$ and the detection parameters Θ to maximize detection accuracy subject to a constraint on the average fronthaul load. We formulate this as a mean squared error (MSE) minimization problem with a bit-budget penalty:

$$\min_{\Theta, \{b_{lk}\}} \quad \mathbb{E}\left[\|\mathbf{s} - f_\Theta(\dot{\mathbf{s}})\|^2\right] \quad \text{s.t.} \quad \mathbb{E}\left[\frac{1}{K}\sum_{l=1}^{L}\sum_{k=1}^{K} b_{lk}\right] \le C_{\text{avg}}, \tag{5}$$

where the expectation is taken over the channel realizations and noise statistics, and C_{avg} represents the target average bits per user per channel use. This formulation implies a task-oriented design: the quantization bit-widths b_{lk} should be dynamically adapted based on the instantaneous local CSI to prioritize users and APs that contribute most significantly to the global detection task, rather than simply minimizing local reconstruction errors.

III. Proposed Joint Quantization-Aware Training Framework

This section details the proposed task-oriented communication framework, designed to jointly optimize bit-width allocation at the Access Points (APs) and signal detection at the Central Processing Unit (CPU). As illustrated in Fig. 1, the architecture comprises two primary components: a distributed *Resource Inference Module* at each AP and a centralized *Bit-Aware Graph Neural Network (GNN)* at the CPU. The processing pipeline operates as follows: First, each AP performs local linear processing to obtain soft estimates of user symbols. Second, a resource-efficient Policy Network at each AP analyzes the reliability of these estimates to infer a quantization bit-width probability distribution. Third, a Differentiable Quantizer generates a quantized message using a soft relaxation of discrete bit-widths to facilitate gradient backpropagation. Finally, the CPU aggregates these heterogeneous-precision signals via a GNN that weights inputs according to their quantization fidelity.

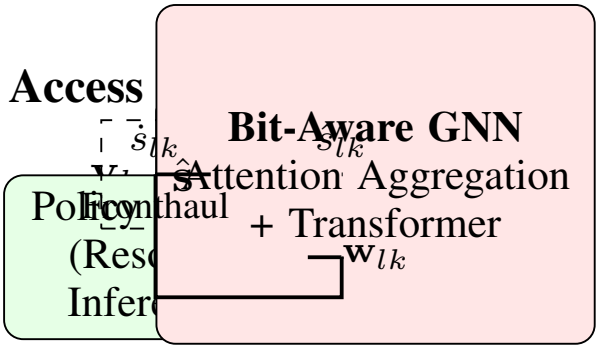

Fig. 1. The proposed Joint Quantization-Aware Training (QAT) framework. Each AP locally infers the optimal bit-width based on instantaneous channel conditions, while the CPU aggregates quantized signals using a bit-aware attention mechanism to compensate for quantization noise.

A. AP-Side Resource Inference Module

To enable dynamic fronthaul adaptation, each AP l is equipped with a Policy Network π_ϕ, parameterized by ϕ, which determines the optimal quantization precision for each user k. Let $\hat{s}_{lk}$ denote the local Linear Minimum Mean Square Error (LMMSE) estimate derived in the system model. The input feature vector to the policy network for the k-th user at AP l is defined as $\mathbf{x}_{lk} = [\Re(\hat{s}_{lk}), \Im(\hat{s}_{lk}), \gamma_{lk}]^T \in \mathbb{R}^3$, where $\gamma_{lk} = \|\mathbf{h}_{lk}\|^2/\sigma^2$ represents the local instantaneous Signal-to-Noise Ratio (SNR). The inclusion of γ_{lk} enables the network to differentiate between noise-dominated observations and informative signal components.

The policy network is implemented as a lightweight Multi-Layer Perceptron (MLP) comprising two fully connected layers with ReLU activation. It maps $\mathbf{x}_{lk}$ to a set of unnormalized logits $\mathbf{z}_{lk} \in \mathbb{R}^{|\mathcal{B}|}$, corresponding to the discrete bit-width set $\mathcal{B} = \{0, 2, 4\}$. To facilitate end-to-end training through the discrete selection process, the Gumbel-Softmax relaxation is employed. During the forward pass, a soft selection vector $\mathbf{w}_{lk} \in [0,1]^{|\mathcal{B}|}$ is generated as:

$$w_{lk,i} = \frac{\exp((\log z_{lk,i} + g_i)/\tau)}{\sum_{j=1}^{|\mathcal{B}|} \exp((\log z_{lk,j} + g_j)/\tau)}, \quad i \in \{1, \ldots, |\mathcal{B}|\}, \tag{6}$$

where g_i are independent samples drawn from the Gumbel$(0,1)$ distribution, and $\tau > 0$ is the temperature parameter. As $\tau \to 0$, $\mathbf{w}_{lk}$ approaches a one-hot vector representing a discrete choice. This mechanism allows gradients to propagate through the categorical decision of bit-widths during the training phase.

B. Differentiable LSQ Quantizer

Upon obtaining the selection weights $\mathbf{w}_{lk}$, the signal $\hat{s}_{lk}$ is processed by a bank of quantizers. The Learned Step Size Quantization (LSQ) method is adopted to minimize quantization distortion for non-Gaussian signals. For a target bit-width $b \in \mathcal{B} \setminus \{0\}$, the quantizer $Q_b(\cdot)$ is defined by a learnable step size parameter Δ_b. The quantization of a scalar value v (representing either the real or imaginary part of $\hat{s}_{lk}$) is expressed as:

$$Q_b(v; \Delta_b) = \Delta_b \cdot \text{round}\left(\text{clip}\left(\frac{v}{\Delta_b}, v_{\min}, v_{\max}\right)\right), \tag{7}$$

where $v_{\min} = -2^{b-1}$ and $v_{\max} = 2^{b-1} - 1$. To enable gradient descent optimization for the step size Δ_b and the input v, the Straight-Through Estimator (STE) is employed during the backward pass. Specifically, the gradient through the rounding operation is approximated as an identity function within the clipping range.

The final transmitted signal $\dot{s}_{lk}$ is a probability-weighted sum of the outputs from the candidate quantizers:

$$\dot{s}_{lk} = w_{lk,0} \cdot 0 + w_{lk,1} \cdot Q_2(\hat{s}_{lk}) + w_{lk,2} \cdot Q_4(\hat{s}_{lk}). \tag{8}$$

This soft composition allows the system to explore the trade-off between transmitting 0 bits (suppression), 2 bits (coarse quantization), and 4 bits (fine quantization) in a differentiable manner.

C. CPU-Side Bit-Aware GNN Detector

At the CPU, the aggregated signals exhibit heterogeneous reliability due to the varying quantization precision applied across different APs. To robustly detect user symbols under these conditions, a Bit-Aware GNN is proposed. A fully connected graph is constructed where the K users function as nodes. The feature vector for the k-th user node aggregates information from all L APs. Standard aggregation methods (e.g., mean pooling) are suboptimal when signal quality varies significantly.

To address this, a *Bit-Aware Attention* mechanism is introduced. The input feature from AP l for user k is augmented with the expected bit-width $\bar{b}_{lk} = \sum_j w_{lk,j} \cdot b_j$. The augmented feature vector is given by $\mathbf{f}_{lk} = [\Re(\dot{s}_{lk}), \Im(\dot{s}_{lk}), \gamma_{lk}, \bar{b}_{lk}]^T$. These features are lifted to a high-dimensional hidden space via an MLP, producing embeddings $\mathbf{h}_{lk}$. The CPU computes

attention weights α_{lk} to determine the significance of the contribution from AP l:

$$\alpha_{lk} = \text{Softmax}_l \left(\text{MLP}_{attn}(\mathbf{h}_{lk}) \right). \tag{9}$$

The inclusion of $\bar{b}_{lk}$ in the input enables the attention network to assign lower weights to APs that transmitted with 0 or 2 bits and higher weights to those with 4 bits, effectively filtering out quantization noise. The aggregated user feature $\mathbf{u}_k = \sum_{l=1}^{L} \alpha_{lk} \mathbf{h}_{lk}$ is subsequently processed by a Transformer-based encoder to mitigate multi-user interference before the final linear projection to the symbol estimate $\hat{s}_k$.

D. Training Strategy and Complexity

The framework undergoes end-to-end training. The objective function is a Lagrangian relaxation that balances detection accuracy and fronthaul capacity:

$$\mathcal{L} = \frac{1}{K} \sum_{k=1}^{K} \|s_k - \hat{s}_k\|^2 + \lambda \left(\frac{1}{LK} \sum_{l,k} \bar{b}_{lk} - C_{\text{avg}} \right)^2, \tag{10}$$

where λ controls the penalty strength for deviating from the target average capacity C_{avg}. An annealing schedule is employed for the Gumbel temperature τ, decaying from 1.0 to 0.1 over the training epochs. This transitions the system from a soft, probabilistic exploration phase to a hard, deterministic selection phase suitable for deployment.

Regarding computational complexity, the AP-side overhead is minimal. The Policy Network requires only $\mathcal{O}(1)$ FLOPs per user-AP pair (specifically, a small MLP with input dimension 3), which is negligible compared to the $\mathcal{O}(N^3)$ complexity of the LMMSE pre-processing. The CPU-side GNN scales linearly with L and quadratically with K due to the attention mechanism, ensuring scalability for dense AP deployments.

IV. Numerical Results

This section presents a numerical evaluation of the proposed Joint Quantization-Aware Training (QAT) framework within a realistic Cell-Free Massive MIMO deployment. We analyze detection accuracy, robustness to Access Point (AP) unavailability, scalability regarding user load, and computational complexity relative to conventional baselines.

A. Simulation Setup

We consider a Cell-Free Massive MIMO system covering a 1×1 km^2 area, simulated using the 3GPP Urban Micro (UMi) channel model with a carrier frequency of 2 GHz and a bandwidth of 20 MHz. The network comprises $L = 16$ distributed APs and $K = 8$ single-antenna User Equipments (UEs) by default, with each AP equipped with $N = 4$ antennas. Large-scale fading incorporates path loss and log-normal shadowing with a standard deviation of 8 dB, while small-scale fading follows a Rayleigh distribution.

We compare the proposed **QAT-GNN** against two baselines:

- **Dist-Full (Ideal)**: APs transmit full-precision (32-bit floating point) local LMMSE estimates to the CPU, serving as the performance upper bound.

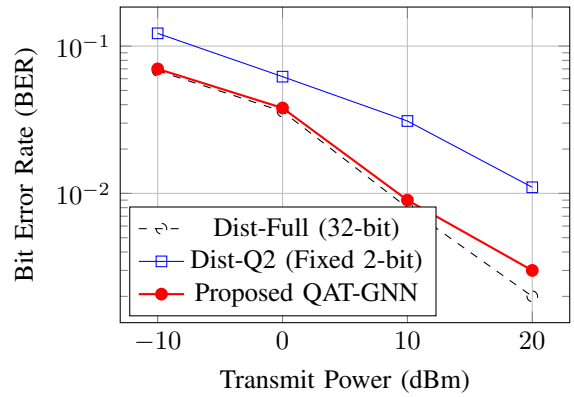

Fig. 2. Bit Error Rate (BER) versus Transmit Power (P_{tx}) for $L = 16$, $N = 4$, $K = 8$. The proposed QAT-GNN achieves performance comparable to the full-precision baseline while significantly reducing fronthaul overhead.

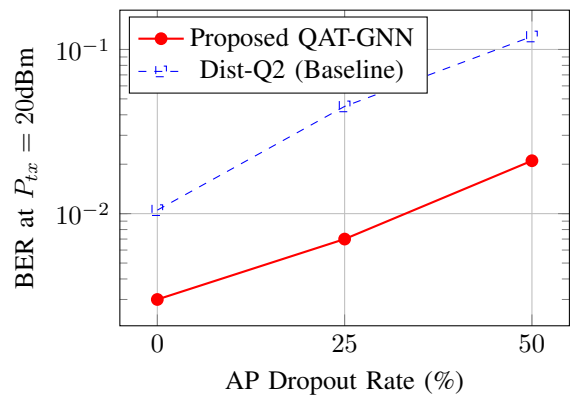

Fig. 3. Robustness to AP Dropout at $P_{tx} = 20$ dBm. The QAT-GNN maintains operational BER levels even when 50% of APs are disconnected, demonstrating superior resilience compared to fixed quantization.

- **Dist-Q2 (Fixed)**: APs transmit 2-bit quantized LMMSE estimates using a fixed Lloyd-Max quantizer, representing a standard low-capacity fronthaul solution.

The proposed QAT-GNN is trained with a target average capacity constraint of $C_{\text{avg}} = 2.5$ bits per symbol.

B. Detection Performance Analysis

Fig. 2 illustrates the Bit Error Rate (BER) performance as a function of user transmit power. The proposed QAT-GNN exhibits a performance profile that closely approximates the ideal Dist-Full baseline across the entire Signal-to-Noise Ratio (SNR) regime. At a transmit power of 20 dBm, the QAT-GNN achieves a BER of approximately 0.3%, comparable to the 0.2% achieved by the full-precision system. In contrast, the Dist-Q2 scheme exhibits an error floor at high SNR, saturating at a BER of 1.1%. This limitation arises because fixed quantization fails to adapt to the high dynamic range of signals in strong channel conditions, resulting in significant quantization distortion. The QAT-GNN mitigates this by dynamically allocating up to 4 bits to high-quality signal components while suppressing noise-dominated streams, thereby maximizing detection fidelity within the bit budget.

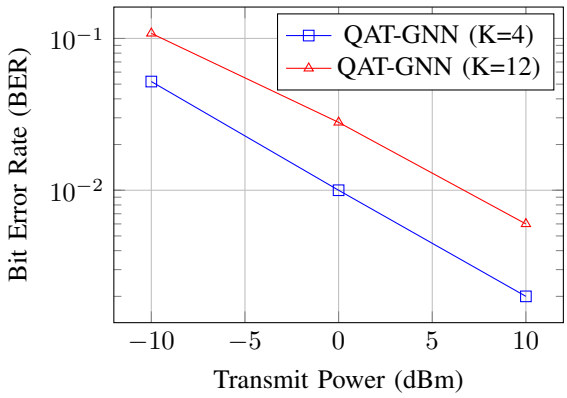

Fig. 4. Impact of User Load on BER performance. The proposed method scales effectively from light load ($K = 4$) to heavy load ($K = 12$), maintaining low error rates even in interference-limited regimes.

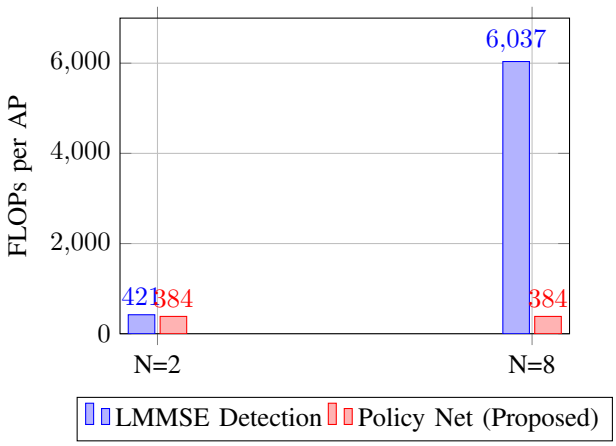

Fig. 5. Computational Complexity Analysis (FLOPs per AP) versus number of antennas N. The proposed Policy Network adds negligible overhead compared to the cubic scaling of LMMSE detection.

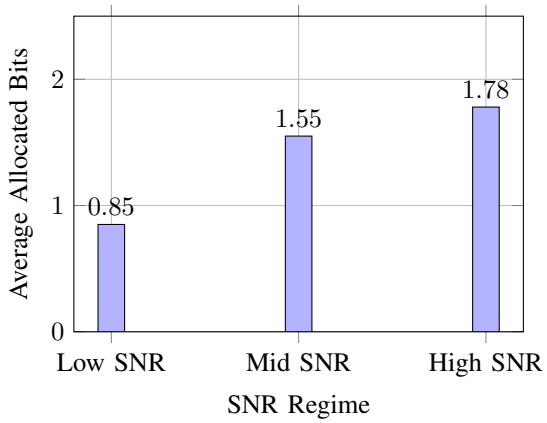

Fig. 6. Task-Oriented Bit Allocation: Average bits allocated per symbol across different SNR regimes. The network autonomously reduces bit consumption in low SNR conditions to conserve bandwidth.

C. Robustness to Infrastructure Failures

A critical advantage of the proposed framework is its resilience to AP unavailability, which may occur due to backhaul outages or power-saving mechanisms. Fig. 3 plots the BER performance against the AP dropout rate, where a subset of APs is randomly silenced. The QAT-GNN demonstrates significant robustness; even with 50% of APs disconnected, the system maintains a BER of approximately 2.1%, which remains within the operational range for standard channel coding rates. Conversely, the Dist-Q2 baseline experiences severe degradation, exceeding 10% BER under identical conditions. This resilience is attributed to the Bit-Aware GNN at the CPU, which leverages the spatial correlation of the remaining active APs to compensate for missing information via the graph-based attention mechanism.

D. Scalability and Complexity

We further investigate system scalability with respect to user density in Fig. 4. Comparing the light load scenario ($K = 4$) with the heavy load scenario ($K = 12$), the QAT-GNN maintains a consistent performance gain over fixed quantization. Notably, in the interference-limited regime ($K = 12$), the non-linear processing capability of the GNN allows it to mitigate multi-user interference more effectively than linear baselines, maintaining a BER below 1% at 10 dBm.

Fig. 5 analyzes the computational overhead per AP. The proposed Policy Network, responsible for determining quantization bit-widths, imposes a constant complexity of 384 FLOPs regardless of the number of antennas N. In contrast, local LMMSE processing scales as $\mathcal{O}(N^3)$. For a system with $N = 8$ antennas, the Policy Network constitutes less than 7% of the total local processing load. This confirms that the proposed intelligent bit-allocation mechanism is computationally efficient and scalable for massive MIMO deployments.

E. Task-Oriented Resource Allocation

Finally, Fig. 6 validates the task-oriented nature of the resource inference module. The figure displays the average number of bits allocated per symbol across different SNR regimes. The network learns to allocate fewer bits (approximately 0.85 bits) in the low SNR regime compared to the high SNR regime (1.78 bits). This adaptive behavior aligns with information-theoretic principles: when the signal is dominated by noise, high-precision quantization captures primarily noise variance, offering diminishing returns for detection. By suppressing these noisy signals (allocating 0 bits) or employing coarse quantization, the system conserves fronthaul capacity for APs with superior channel conditions, thereby optimizing the global detection objective.

References

[1] J. Lewandowsky, S. Dongare, R. M. Lima, M. Adrat, M. Schrammen, and P. Jax, "Genetic algorithms to maximize the relevant mutual information in communication receivers," *Electronics*, 2021.

[2] C. Jeon, R. Ghods, A. Maleki, and C. Studer, "Optimality of large mimo detection via approximate message passing," *Unknown*, 2015.

[3] P. Som, T. Datta, A. Chockalingam, and B. S. Rajan, "Improved large-mimo detection based on damped belief propagation," *Unknown*, 2010.

[4] H. He, C.-K. Wen, S. Jin, and G. Y. Li, "Model-driven deep learning for mimo detection," *IEEE Transactions on Signal Processing*, 2020.

[5] Y. Wei, M. Hong, M. Zhao, M. Lei, and M. Lei, "Learned conjugate gradient descent network for massive mimo detection," *IEEE Transactions on Signal Processing*, 2020.

[6] J. Sun, Y. Zhang, J. Xue, and Z. Xu, "Learning to search for mimo detection," *IEEE Transactions on Wireless Communications*, 2020.
[7] J. Shao, Y. Mao, and J. Zhang, "Task-oriented communication for multi-device cooperative edge inference," *arXiv (Cornell University)*, 2021.
[8] ——, "Task-oriented communication for multi-device cooperative edge inference," *arXiv preprint arXiv:2109.02011*, 2021.
[9] E. Beck, C. Bockelmann, and A. Dekorsy, "Semantic information recovery in wireless networks," *Sensors*, 2023.
[10] S. Li, Y. Wang, S. Guo, and C. Feng, "Task-oriented communication for graph data: A graph information bottleneck approach," *arXiv preprint arXiv:2401.01234*, 2024.
[11] G. He, S. Cui, Y. Dai, and T. Jiang, "Learning task-oriented channel allocation for multi-agent communication," *IEEE Transactions on Vehicular Technology*, 2022.
[12] H. Hou, S. Bi, X. Li, S. Wang, L. Qian, and Z. Quan, "Scalable multi-task edge sensing via task-oriented joint information gathering and broadcast," *arXiv preprint arXiv:2501.01234*, 2025.